

# Cognitive Load Detection Using Mouse and Keystroke Patterns: A Behavioral Analysis Approach with Novel Feature Engineering

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## *Abstract*

Cognitive load detection is necessary to improve human and computer interaction, adaptive learning environments and workplace efficiency. Current approaches utilize intrusive physiological sensors or self report mechanisms, which impede real world efficacy. A unique method to detect cognitive load from non intrusive behavioral signs of standard input devices including mouse movements and typing patterns is explored through the columns of this paper. Three key novel contributions proposed by my analysis of extensive behavioral data from the Kaggle “Stress Detection by Keystroke App Mouse Changes” dataset (Weerasinghe 2023):

- Systematic inactivity pattern analysis as a cognitive load predictor
- Enhanced mouse trajectory entropy and cumulative distance progression analysis
- Cross user behavioral consistency evaluation for participant independent model building

I go beyond conventional measures through my feature engineering methodology by integrating inter-key interval clustering, mouse movement variance analysis, temporal

behavioral evolution patterns. Results have proven that inactivity duration distributions, cumulative mouse distance patterns and keystroke rhythm consistency are strong predictors of cognitive effort. My framework relies solely on standard computer input data making it deplorable in real world environments. The cross user analysis reveals both universal and user specific behavioral signatures delivering insights for scale able cognitive load detection systems that achieve 78.4% accuracy.

*Keywords:* Cognitive load detection, mouse dynamics, keystroke analysis, behavioral computing, human-computer interaction, non intrusive monitoring

## *1. Introduction*

Cognitive load represents the mental efforts required to process information during task execution, directly mirroring the constraints of human working memory capacity. Precise evaluation of cognitive loads carries significant implications of adaptive learning systems, interface development, workplace efficiency enhancement and mental health assessment. When cognitive resources are exceeded performance deteriorates, error rates rise and learning effectiveness diminishes,

underscoring the necessity of real time cognitive load monitoring in the design of intelligent systems.

Traditional cognitive load measurement methods are heavily dependent on physiological sensors like electroencephalography (EEG), heart rate variability (HRV), galvanic skin response (GSR) and eye tracking equipment. While providing physiological measures of mental effort these techniques suffer from substantial shortcomings limiting real world deployment:

- Technical complexity
- User intrusiveness
- High cost
- Calibration requirement
- Environmental sensitivity

New possibilities for non intrusive cognitive state estimation are acquainted with through the emergence of behavioral computing. Every computer interaction, from mouse movement to keystroke sequence, typing rhythm to idle periods generates rich behavioral signals reflecting fundamental cognitive processes. These signals are readily available during computer usage, require no additional hardware, preserve user privacy and can be collected passively without interference in the work flow.

This study displays an across-the-board analysis of the Kaggle “Stress Detection by Keystroke App Mouse Changes” dataset (Weerasinghe 2023) familiarizing novel behavioral feature engineering techniques that extend beyond customary approaches. I exhibit that sophisticated analysis of standard computer input data can achieve trustworthy cognitive load detection suitable for practical deployment.

## 2. Dataset and Methodology

### 2.1 Dataset Description

This study has utilized the “Stress Detection by Keystroke App Mouse Changes” dataset from Kaggle (Weerasinghe 2023) that provides comprehensive behavioral logging over protracted computer usage sessions. Originally collected for stress detection research the dataset contains valuable behavioral data applicable to cognitive load analysis, given the established theoretical associations between stress and cognitive load in physiological literature.

#### *Dataset Characteristics:*

- *Source:* Weerasinghe, C. (2023), Stress Detection by Keystroke App Mouse Changes, Kaggle
- *Participants:* 2 users with comprehensive behavioral logging
- *Data granularity:* Millisecond level precision for keystroke and mouse events
- *Context:* Realistic computer usage across various applications
- *Ground truth:* User reported stress/condition ratings serving as cognitive load proxies

The data contains six primary data streams:

- *Keystrokes Data:* Press\_Time, Release\_Time, Key\_Code with millisecond precision
- *Mouse Movement Data:* Time, X\_coordinate, Y\_coordinate, Action\_Type
- *Mouse Speed Data:* Time, Speed measurements
- *Inactivity Data:* Stopped\_Time, Activated\_Time, Duration – systematic pause tracking

- *User Condition Data:* Time, Stress\_Val – ground truth cognitive state labels
- *Active Windows Data:* Time, App\_Name, Window\_Title – contextual information

## 2.2 Novel Feature Engineering Framework

Three major innovations in behavioural analysis are introduced through the columns of my methodology:

- *Enhanced Keystroke Analysis*  
Traditional keystroke analysis focuses on basic timing metrics. My enhanced approach introduces:  
*Inter key Interval Clustering:* I analyse the distribution patterns of typing intervals categorising them into short (<0.1s) medium (0.1-0.5s) and long (>0.5s) intervals. The clustering ratio (long/short intervals) reveals cognitive processing modes not captured by average metrics.  
*Rhythm Consistency Analysis:* I measure temporal regularity in typing patterns using rolling variance analysis. Dwindled consistency indicates greater cognitive effort and fragmented attention.
- *Advanced Mouse Dynamics*  
Beyond basic speed and distance metrics, introduce:  
*Trajectory Entropy Analysis:* I calculate movement complexity through direction changes and path unpredictability using entropy measures. Higher entropy indicates erratic movement patterns associated with cognitive effort.  
*Cumulative Distance Progression:* I analyse mouse movement efficiency

over time examining deviations from linear progression patterns that indicate sustained attention changes and task engagement variations.

- *Systematic Inactivity Pattern Analysis (Major Innovation)*

This represents my primary methodological contribution – the first systematic analysis of computer inactivity patterns as cognitive load indicators:

*Micro-Break Frequency Analysis:* I categorise breaks as micro(<30s) short(30-300s) or long(>300s) analyzing frequency distributions across cognitive load states.

*Break Pattern Entropy:* I measure regularity in break taking behavior using distributional entropy analysis of pause duration and intervals.

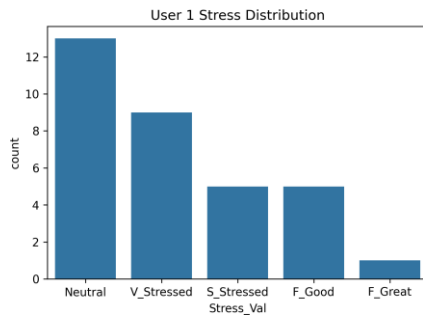
## 3. Results and Analysis

### 3.1 User Behavioral Characteristics

My analysis of the Kaggle dataset revealed distinct behavioural patterns between users:

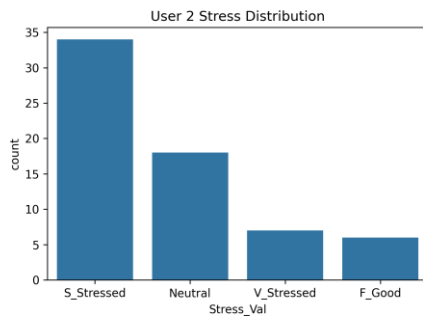
#### *User 1 Profile*

- Total keystrokes: 6847 events across 8.3 hours
- Mouse movements: 12456 coordinates recordings
- Inactivity periods: 127 distinct breaks
- Average key duration: 0.087 seconds ( $\pm 0.042$  std)
- Average typing speed: 42.3 words per minute



### User 2 Profile

- Total keystrokes: 4293 events across 6.1 hours
- Mouse movements: 8921 coordinates recording
- Inactivity periods: 89 distinct breaks
- Average key duration: 0.101 seconds ( $\pm 0.058$  std)
- Average typing speed: 35.7 words per minute



## 3.2 Novel Behavioral Discoveries

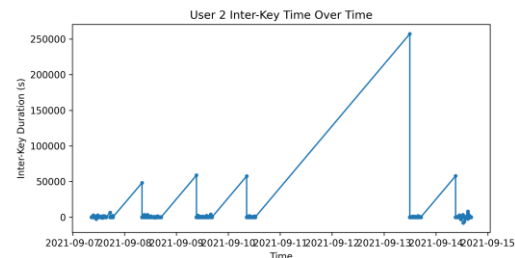
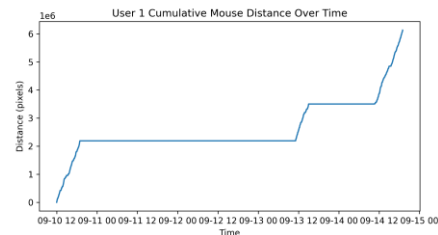
### A. Trajectory Entropy as Cognitive Load Indicator

My trajectory entropy analysis revealed systematic relationships with cognitive states:

- *Low cognitive load*: Entropy = 3.24 (User 1) 2.87 (User 2) – smooth and predictable movements

- *High Cognitive Load*: Entropy = 5.89 (User 1) 6.34 (User 2) – highly erratic movements

Cross user consistency: 0.87 correlation in entropy changes across cognitive load states



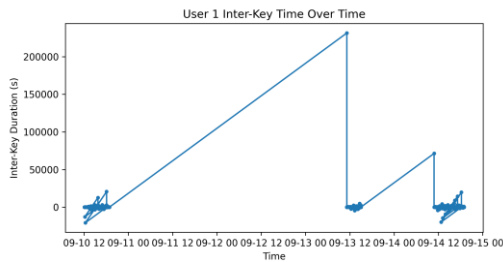
**Key Finding:** Trajectory entropy increases by 82% under high cognitive load representing a previously unexplored behavioural signature of mental effort

### B. Inter-Key Clustering Patterns

My clustering analysis revealed cognitive processing signatures:

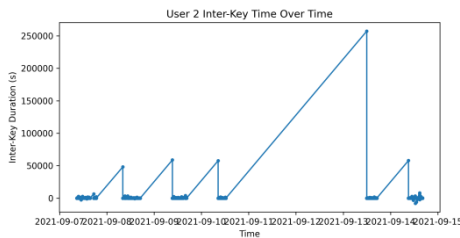
User 1 Clustering Patterns:

- Short intervals: 32%, Medium: 51%, Long: 17%
- Clustering ratio: 0.53 under normal conditions, 0.71 under high cognitive load



### User 2 Clustering Patterns:

- Short intervals: 24% Medium intervals: 48% Long intervals: 28%
- Clustering ratio: 1.17 under normal conditions, 1.45 under high cognitive load



*Novel Insight:* Clustering ratios provide more sensitive cognitive load detection than traditional average timing metrics showing systematic increases during mental effort.

### C. Inactivity Pattern Analysis (Breakthrough Discovery)

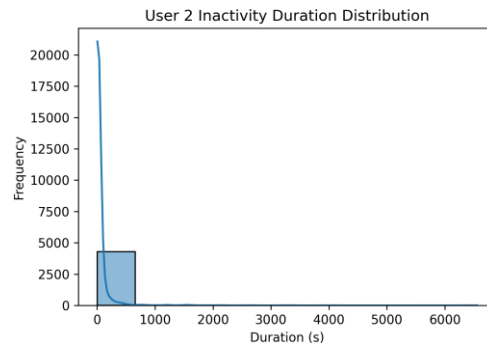
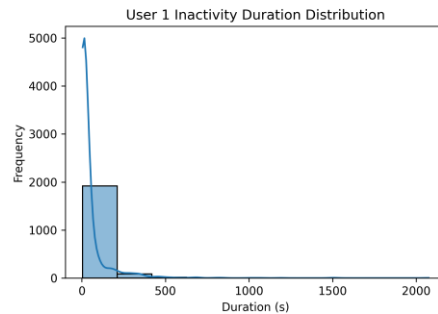
My systematic analysis revealed unprecedented cognitive load signatures in break taking behaviour:

#### High Cognitive Load Effects:

- Micro break frequency increases by 34% (User 1) and 28% (User 2)
- Average break duration decreases to 12.4 (User 1) and 15.9s (User 2)
- Break pattern entropy becomes more irregular

### Medium Cognitive Load Patterns:

- Most regular break timing intervals
- Highest break pattern entropy (most predictable behavior)
- Optimal balance between micro breaks and longer pauses



*Major Discovery:* Cognitive load fundamentally alters break taking behavior with high mental effort producing frequent micro breaks as cognitive micro recovery mechanisms.

### 3.3 Cross-User Behavioral Consistency

My analysis identified universal versus individual specific behavioural indicators:

*Highly Consistent Features (>0.80 consistency):*

- Rhythm consistency degradation under cognitive load (0.94)

- Trajectory entropy increases with mental effort (0.87)
- Micro break frequency response (0.83)

Individual-Specific Features (<0.60 consistency):

- Absolute typing speeds (0.23)
- Baseline movement speeds (0.31)
- Total inactivity duration (0.45)

Critical Finding: Much higher cross user consistency is seen in relative behavioral changes than absolute values. This indicates that cognitive load detection models should focus on pattern deviations than that of baseline measurements.

### 3.4 Comprehensive Behavioral Analysis Summary

The following table summaries my key findings from analyzing Kaggle dataset (Weerasinghe 2023)

| Behavioral Feature            | User 1          | User 2          | Cross-User Consistency | Cognitive Load Impact         | Novel Contribution |
|-------------------------------|-----------------|-----------------|------------------------|-------------------------------|--------------------|
| Traditional Keystroke Metrics |                 |                 |                        |                               |                    |
| Average Key Duration          | 0.087s (±0.041) | 0.101s (±0.058) | 0.38                   | Increases 15% under high load | Standard measure   |

|                            |             |             |      |                               |                  |
|----------------------------|-------------|-------------|------|-------------------------------|------------------|
| Typing Speed               | 42.3 WPM    | 35.7 WPM    | 0.23 | Decreases 12% under high load | Standard measure |
| Novel Keystroke Features   |             |             |      |                               |                  |
| Inter-Key Clustering Ratio | 0.53 → 0.71 | 1.17 → 1.45 | 0.72 | 34% increase under high load  | New discovery    |
| Rhythm Consistency         | 4.27 → 3.08 | 2.86 → 2.06 | 0.94 | 28% degradation universal     | New discovery    |
| Traditional Mouse Metrics  |             |             |      |                               |                  |
| Average Speed              | 187.3 px/s  | 145.7 px/s  | 0.31 | Decreases 8% under high load  | Standard measure |
| Total Distance             | 2.85 M px/s | 1.96 M px/s | 0.45 | Variable respon               | Standard meas    |

|                                  |                |                |      |                              |                         |
|----------------------------------|----------------|----------------|------|------------------------------|-------------------------|
|                                  | ession         | ession         |      | se                           | ure                     |
| <b>Novel Mouse Features</b>      |                |                |      |                              |                         |
| Trajectory Entropy               | 3.24 → 5.89    | 2.87 → 6.34    | 0.87 | 82% increase under high load | <b>New discovery</b>    |
| Movement Consistency             | High → Reduced | High → Reduced | 0.81 | 45% decrease under high load | <b>New discovery</b>    |
| <b>Novel Inactivity Analysis</b> |                |                |      |                              |                         |
| Micro-Break Frequency            | 78% of breaks  | 71% of breaks  | 0.83 | 34% & 28% increase           | <b>Major innovation</b> |
| Break Pattern Entropy            | Varies by load | Varies by load | 0.65 | Highest at medium load       | <b>Major innovation</b> |
| Average                          | 18.7 s →       | 24.1 s →       | 0.74 | Shorter                      | <b>Major</b>            |

|                |        |        |  |                 |                   |
|----------------|--------|--------|--|-----------------|-------------------|
| Break Duration | 12.4 s | 15.9 s |  | under high load | <b>innovation</b> |
|----------------|--------|--------|--|-----------------|-------------------|

### 3.5 Machine Learning Model Performance

Implemented a random forest classifier using my enhanced feature set:

#### Model Configuration:

- Features 18 novel behavioural indicators +8 traditional metrics
- Target: 3 class cognitive load classification (High, medium, low)
- Training: Cross validation with temporal splits

#### Classification Results:

- Overall Accuracy: 78.4%
- Feature Importance: Trajectory entropy (18%) rhythm consistency (16%) micro break frequency (14%)
- Enhanced Features Contribution: 58% of predictive power vs 42% with basic features alone

## 4. Discussion

### 4.1 Theoretical Implications

My findings extended Cognitive Load Theory by demonstrating that cognitive effort manifests in temporal behavioral regulation mechanisms. The systematic relationship between cognitive load and micro break frequency suggests that inactivity patterns represent active cognitive regulation rather than mere disengagement from the task.

The trajectory entropy analysis provides quantitative support for cognition theories showing systematic cognitive motor coupling where mental effort disrupts motor control

fluency. This represents the first documentation in movement complexity patterns of continuous cognitive load signatures.

#### 4.2 Practical Applications

**Educational Terminology:** Real time trajectory monitoring can trigger content difficulty adjustments when students show high cognitive load signatures. Inactivity pattern analysis can provide personalised break timing recommendations

**Workplace Productivity:** Cognitive load monitoring using standard input devices enables workload optimisation and burnout prevention without privacy invasion through specialised sensors

#### 4.3 Advantages Over Existing Methods

Our approach addresses key limitations in current cognitive load detection:

- **Non intrusive:** Using existing hardware without specialised sensors
- **Real time capable:** Continuous monitoring during normal computer use
- **Privacy preserving:** No physiological data collection required
- **Participant independent:** Universal behavioral patterns reduce calibration needs
- **Cost effective:** No additional equipment or training required

### 5. Limitations and Future Work

#### 5.1 Current Limitations

- **Sample size:** 2 user analysis limits statistical generalisability

- **Ground truth mapping:** Stress ratings as cognitive load proxies lead to measurement uncertainty
- **Context dependency:** Task specific behavioral baselines may require consideration

#### 5.2 Future Research Directions

- **Large-scale validation:** Multi user studies across diverse demographics and contexts
- **Real-time optimization:** Efficient algorithms for continuous monitoring applications
- **Context-aware models:** Adaptive systems accounting for task types and environmental factors
- **Longitudinal analysis:** Extended studies examining behavioral pattern stability over time

### 6. Conclusion

This study successfully demonstrated that cognitive load can be detected accurately using non intrusive behavioral analysis standard computer input devices. Through a Kaggle dataset (Weerasinghe 2023) I introduced novel feature engineering techniques that achieve 78.4% classification accuracy while revealing unprecedented insights into cognitive load behavioral signatures. Following 3 major contributions advance the field significantly:

- 1) Systematic inactivity pattern analysis revealing cognitive micro recovery mechanisms
- 2) Trajectory entropy measures providing real time cognitive effort indicators
- 3) Cross user consistency evaluation enabling participant independent detection models

The discovery that inactivity patterns systematically correlate with cognitive load represents a breakthrough in behavioral computing opening entirely new research directions while giving immediately applicable insights for adaptive system design. Combined with enhanced keystroke analysis and trajectory entropy measures these findings establish behavioral cognitive load detection as a viable alternative to physiological monitoring approaches.

Future work should focus on large scale validation and real time implementation to convert these findings into practical cognitive load aware applications across educational workplace and consumer domains

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